

46 YEARS OF EXCELLENCE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-18(FN)** **Date:16.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In an experiment the angles are required to be measured using an instrument similar to vernier calliper. 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half a-degree ($=0.5^0$), then the least count of the instruement is
A) one minute B) half minute C) one degree D) half degree

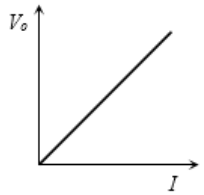
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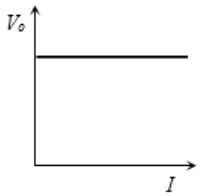
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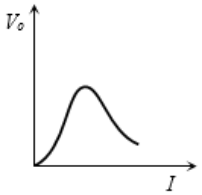
2. A block of mass 1 kg slides down on a rough inclined plane of inclination 60° starting from its top. If the coefficient of kinetic friction is 0.5 and length of the plane is 1 m, then work done against friction till it reaches bottom is (Take $g = 9.8\ m/s^2$)

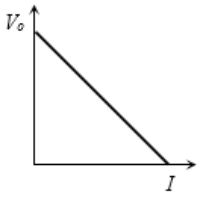
A) 9.82 J B)4.94 J C) 2.45J D) 1.96 J

3. The correct curve between the stopping potential (V) and intensity of incident light (I) is

A) 

B) 

C) 

D) 

4. Ice with mass $M_1 = 600\ g$ at temperature $t_1 = -10^0 C$ is placed into a copper vessel heated to $t_2 = 350^0 C$. As a result, the vessel now contains $M_2 = 550g$ of ice mixed with water. The specific heat of copper = $0.1\ cal/g^0C$ and specific heat of ice = $0.5\ cal/g^0C$. Latent heat of ice = 80 cal/g. The mass of vessel is $M \times 100\ g$. The value of M is

A) 2 B) 3 C) 4 D) 5

5. In the nuclear fusion reaction ${}_1H^2 + {}_1H^3 \rightarrow {}_2He^4 + n$ given that the repulsive potential energy between the two nuclei is $7.7 \times 10^{-14} J$, the temperature at which the gases must be heated to initiate the reaction is nearly [Boltzmann's constant $k = 1.38 \times 10^{-23} J/K$].

A) 10^7 K B) 3.7×10^5 K C) 10^3 K D) 3.7×10^9 K

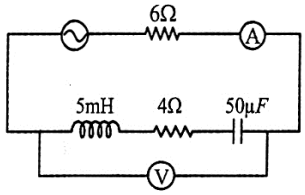
6. A monoatomic gas is suddenly compressed to (1/8)th of its initial volume adiabatically. The ratio of its final pressure to the initial pressure is (given the ratio of the specific heats of the given gas to be 5/3)

A) 32 B) 40/3 C) 24/5 D) 8

7. What is the angular momentum of an electron in Bohr's hydrogen atom whose energy is – 0.544 eV.

A) $\frac{h}{\pi}$ B) $\frac{2h}{\pi}$ C) $\frac{5h}{2\pi}$ D) $\frac{7h}{2\pi}$

8. In the circuit shown in the figure, the ac source gives a voltage V = 20 cos (2000t). Neglecting source resistance, the voltmeter and ammeter reading will be



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A) 0V, 0.97 A B) 1.68 V, 0.47 A C) 0V, 1.4 A D) 5.64 V, 1.4 A

9. A block of mass m is placed on a surface with a vertical cross section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is

A) $\frac{1}{2}$m B) $\frac{1}{6}$m C) $\frac{2}{3}$m D) $\frac{1}{3}$m

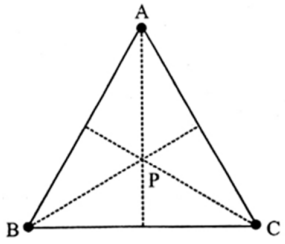
10. The radiation of energy E falls normally on a perfectly reflecting surface. The momentum transferred to the surface is

A) $\frac{E}{C}$ B) $\frac{2E}{C}$ C) EC D) $\frac{E}{C^2}$

11. A beam of ordinary light is incident on a system of 4 polariods which are arranged in succession such that each polariod is turned through $30°$ with respect to the preceding one. The percentage of the incident intensity that emerges out from the system is approximately

A) 21 % B) 6.25 % C) 28 % D) 14 %

12. Three equal masses (each m) are placed at the corners of an equilateral triangle of side 'a'. Then the escape velocity of an object from the circumcentre P of triangle is



A) $\sqrt{\frac{2\sqrt{3}Gm}{a}}$ B) $\sqrt{\frac{\sqrt{3}Gm}{a}}$ C) $\sqrt{\frac{6\sqrt{3}Gm}{a}}$ D) $\sqrt{\frac{3\sqrt{3}Gm}{a}}$

13. A particle of charge $-q$ and mass m moves in a circular orbit of radius r about a fixed charge +Q, the relation between the radius of the orbit r and the time period T is

A) $r = \frac{Qq}{16\pi^2 \in_0 m} T^2$ B) $r^3 = \frac{Qq}{16\pi^3 \in_0 m} T^2$

C) $r^2 = \frac{Qq}{16\pi^3 \in_0 m} T^3$ D) $r^2 = \frac{Qq}{4\pi^3 \in_0 m} T^3$

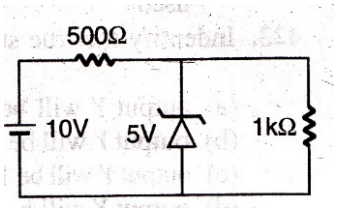
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14. Using mass [M], length [L], time [T] and current [A] as fundamental quantities, the dimensional formula of permittivity is

A) $ML^{-2}T^{2}A$ B) $M^{-1}L^{-3}T^{4}A^{2}$ C) $MLT^{-2}A$ D) $ML^{2}T^{-1}A^{2}$

15. In the following circuit, the current flowing through $1k\Omega$ resistor is :

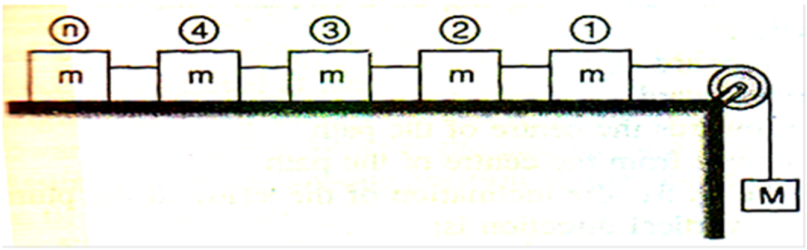


A) 0 mA B) 5 mA C) 10 mA D) 15 mA

16. If 2 moles of mono atomic gas (γ=5/3) and one mole of diatomic gas (γ=7/5) are mixed then γ of the mixture is

A) 23/15 B) 17/11 C) 3/2 D) 71/45

17. In the given arrangement, n number of equal masses are connected by strings of negligible masses. The tension in the string connected to nth mass is



A) $\frac{mMg}{nm+M}$ B) $\frac{mMg}{nmM}$ C) mg D) mng

18. 3 identical bulbs are connected in series and these together dissipate a power P. If now the bulbs are connected in parallel, then the power dissipated will be

A) $\frac{P}{3}$ B) $3P$ C)$9P$ D) $\frac{P}{9}$

19. Which of the following electromagnetic waves has the highest frequency?

A) Radio waves B) X-rays C) γ -rays D) Micro waves

20. A ball is thrown from a point with a speed v_0 at an angle of projection θ. From the same point and at the same instant a person starts running with a constant speed $\frac{v_0}{2}$ to catch the ball, will the person be able to catch the ball. If yes, what should be the angle of projection.

A) yes, 60^0 B) yes, 30^0 C) No D) yes, 45^0

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. In a transformer, number of turns in the primary coil are 140 and that in the secondary coil are 280. If current in primary coil is 4A then the current (in A) in the secondary coil is

22. A plano convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface, now this lens has been used to form the image of an object. At what distance from the lens an object be placed in order to have a real image of the size of the object [in cm]

23. A steady current I goes through a wire loop PQR having shape of a right angle triangle with $PQ = 3x,\ PR = 4x$ & $QR = 5x$ the magnitude of the magnetic field at P due to this loop is $K\left[\frac{\mu_0 I}{48\pi x}\right]$ find the value of k.

24. A given charge situated at a certain distance from an electric dipole in the end-on position, experiences a force F. If the distance of the charge is doubled, the force acting on the charge will be $\frac{F}{N}$. Then the value of N is

25. An ammeter reads up to 1 ampere, its Internal resistance is 0.81 ohm, to increase the range to 10A the value of the shunt required is $\frac{N}{100}$ ohm, then value of N is ____

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CHEMISTRY **MAX.MARKS: 100**

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26. Number of bridge ligands and co-ordination number in complex compound $\left[Co_2(CO)_8\right]$ is

A) 2,6 B) 3,6 C) 2,4 D) 6,3

27. What is the shape of I_3^- and XeF_2 respectively.

A) LIneear, Angular B) Linear, Linear

C) Angular, Angular D) Angular, Linear

28. Correct statement/s about hydrides of 16th group elements

A) Reducing property increases H_2S To H_2Te

B) Boiling point order is $H_2O > H_2Te > H_2Se > H_2S$

C) Only H_2O is a liquid at room temperature

D) All are correct

29. Select the correct set of species with paramagnetic nature

A) NO, O_2^+, O_2^-, B_2 B) B_2, O_2, N_2, N_2^+

C) $O_2, O_2^+, O_2^{-2}, N_2^+$ D) Be_2, He_2, C_2, N_2

30. A solution containing Na_2CO_3 and $NaOH$ requires 300 ml of $0.1N$ HCl using phenolphthalein as an indicator. Methyl orange is then added to the above titrated solution when a further 25 ml of $0.2N$ HCl is required. The amount (in grams) of $NaOH$ present in solution is

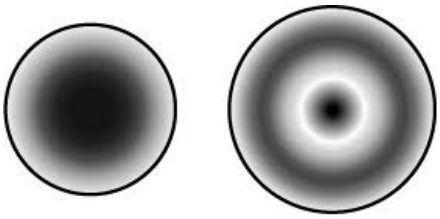
(Molecular weights: $NaOH = 40$, $Na_2CO_3 = 106$)

A) 1g B) 1.5 g C) 2 g D) 4 g

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31. The probability density plots of 1s and 2s orbitals are given in



1s orbital 2s orbital

The density of dots in a region represents the probability density of finding electrons in the region. On the basis of above diagram which of the following statement is INCORRECT ?

A) 1s and 2s orbitals are spherical in shape.

B) The probability of finding the electron is maximum near the nucleus.

C) The probability of finding the electron at a given distance is equal in all directions.

D) The probability density of electrons for 2s orbital increases as the distance from the nucleus increases.

32. A mixture contains 4 grams of $He_{(g)}$, 3.01×10^{22} molecules of $CO_{2(g)}$, 5.6 Lit of $CH_{4(g)}$ (at STP). Then calculate mole fraction of $He_{(g)}$. (Given: $N_0 = 6.02\times10^{23}$, Atomic weight of $He = 4$)

A) $\frac{1}{1.30}$ B) $\frac{0.05}{1.30}$ C) $\frac{1}{0.05}$ D) $\frac{0.25}{1.30}$

33. Select Correct combination of given data

a) Chromate Ion –Yellow colour

b) di choromate Ion– orange colour

c) manganate Ion – green colour

d) MnO_2 – Black colour

A) a, b only B) a, b, c only C)a, d only D) a, b, c, d

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34. Arrange the following in decreasing order of their boiling points.

a) n–butane b) 2–methyl butane

c) n-pentane d) 2,2–dimethyl propane

A) a > b > c > d B) b > c > d > a C) d > c > b > a D) c > b > d > a

35. Oxidation number of chromium in CrO_4^{-2}, $Cr_2O_7^{-2}$, CrO_5 is

A) +6, +6, +6 respectively B) +6, +6, +10 respectively

C) +6, +6, +8 respectively D) +6, +7, +8 respectively

36. Three aqueous solutions of NaCl labelled as A, B, C with concentration 0.1M, 0.01M and 0.001 M respectively. The value of Van't hoff factor for these solutions will be in the order

A) $i_A = i_B = i_C$ B) $i_A > i_B > i_C$ C) $i_A < i_B < i_C$ D) $i_A < i_B > i_C$

37. $\left[Co(NH_3)_2(NO_2)(H_2O)_2Cl\right]Br$ shows what type of isomerism

A) Linkage, ,ionization, hydrated only B) Hydrated, linkage only

C) ionization, hydrated only D) linkage, isomersim only

38. $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$. Rate of reaction is given as $1.2\times10^{-2} M/\sec$ then correct option is

A) with respect to each term rate of reaction is $1.2\times10^{-2} M/\sec$

B) Rate of appearance of NH_3 is $2.4\times10^{-2} M/\sec$

C) rate of disappearance of H_2 is $3.6\times10^{-2} M/\sec$

D) all are correct

39. Choose the correct option for free expansion of an ideal gas under adiabatic condition is

A) $q = 0,\ W = 0,\ \Delta T = 0$ B) $q = 0,\ W = 0,\ \Delta T \neq 0$

C) $q = 0,\ W = 0,\ \Delta T < 0$ D) $q = 0,\ W = 0,\ \Delta T > 0$

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40. Based on given E^0 values select the correct option

E^0 values: $Br_2 / Br^- = 1.90V$, $Ag^+ / Ag = +0.80V$, $Cu^{+2} / Cu = +0.34V$, $I_2 / I^- = 0.54V$

A) Cu will reduce Br^- B) Cu will reduce Ag

C) Cu will reduce I^- D) Cu will reduce Br_2

41. Different fraction of crude oil in petroleum industry are separated by

A) distillation B) Fractional distillation

C) steam distillation D) vacuum distillation

42. Correct option from the given data is

A) Acidic: $H_3PO_2 > H_3PO_3 > H_3PO_4$

B) Basicity: $H_3PO_2 < H_3PO_3 < H_3PO_4$

C) Reducing capacity: $H_3PO_2 > H_3PO_3 > H_3PO_4$

D) All are correct

43. The product (C) in the below mentioned reaction is :

$$CH_3 - CH_2 - CH_2 - Br \xrightarrow[\Delta]{KOH_{(alc)}} \underset{(major)}{A} \xrightarrow{HBr} \underset{(major)}{B} \xrightarrow[KOH_{(aq)}]{\Delta} \underset{(major)}{C}$$

A) Propan-1-ol B) Propene C) Propyne D) Propan-2-ol

44. Cannizaro raction is not given by

A) CCl_3CHO B) $(Ph)_3CCHO$ C) CH_3CHO D) all

45. Number of alkenes that shows geometrical isomerism with formula C_5H_{10}.

A) 4 B) 1 C) 3 D) 2

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46. Total number of reagents from those given below that can convert nitrobenzene into aniline is _____

$I)\ Sn-HCl \quad II)Zn-NH_4OH \quad III)Fe-HCl$
$IV)Zn-HCl \quad V)H_2-Pd \quad VI)H_2-Raney\ Nickel$

47. Number of elements that gives colour in flame test from the following

$Li,\ Na,\ K,\ Rb,\ Be,\ Mg,\ Ca.$

48. At $25^{\circ}C$ the solubility product of $Mg(OH)_2$ is 1.0×10^{-11}. At which pH, will Mg^{2+} ions start precipitating in the form of $Mg(OH)_2$ from a solution of $0.001\,M$ Mg^{2+} ions?

49. Number of Chiral carbons in $\alpha-D-(+)-Gluco\ Pyranose$

50. In the formation of phorone, number of acetone molecules involved is

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MATHS **MAX.MARKS: 100**

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

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51. If the system of linear equations $x+y+z=6, x+2y+3z=14$ and $2x+5y+\lambda z=\mu$ $(\lambda,\mu\in R)$ has a unique solution, then

A) λ cannot be 8 and μ cannot be 36

B) λ must be 8 and μ must be 36

C) λ cannot be 8 and μ can be any real number

D) λ and μ must be 6

52. If $\operatorname{cosec}\theta-\sin\theta=m$ and $\sec\theta-\cos\theta=n$ such that $\left(m^2n\right)^{2/3}+\left(mn^2\right)^{2/3}=k$ then k = ___

A) 2 B)1 C) 3 D) 4

53. Three vectors $\bar{a}\left(|\bar{a}|\neq 0\right), \bar{b}$ and $\bar{c}$ are such that $\bar{a}\times\bar{b}=3\bar{a}\times\bar{c}$, also $|\bar{a}|=|\bar{b}|=1$ and $|\bar{c}|=\frac{1}{3}$. If $\left(\bar{b},\bar{c}\right)=60^0$ and $|\bar{b}-3\bar{c}|=\lambda|\bar{a}|$, then $\sin^{-1}\left(\lambda^2\right)$_____

A) $\frac{\pi}{2}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{4}$ D) $\frac{\pi}{6}$

54. If the length of the semi minor axis of an ellipse is equal to $\frac{1}{3}$ of the distance between the foci, then the eccentricity of the ellipse is

A) $\frac{5}{\sqrt{13}}$ B) $\frac{\sqrt{3}}{2}$ C) $\frac{4}{\sqrt{13}}$ D) $\frac{3}{\sqrt{13}}$

55. If 'M' and 'm' are the local maximum and local minimum values of $f(x)=\frac{40}{3x^4+8x^3-18x^2+60}$, in $x\in(-2,9)$, then $\frac{1}{M}+\frac{1}{m}=$_____

A) $\frac{131}{40}$ B) $\frac{130}{41}$ C) $\frac{113}{40}$ D) $\frac{113}{41}$

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56. A man starts form the point $P(-3,4)$ and reaches the point $Q(0,1)$ touching the $x-axis$ at $R(\alpha,0)$ such that $PR+RQ$ is minimum. Then $|5\alpha|=$ _____

A) 3 B) 1 C) 2 D) 4

57. Let 'L' and 'M' be two distinct positive real numbers . let 11th term of a G.P., whose first term is 'L' and 3rd term is 'M', is equal to K^{th} term of another G.P., whose first term is 'L' and fifth term is 'M', then the number of divisors of 'K', is _____

A) 4 B) 3 C) 2 D) 1

58. Let $\bar{a}$ and $\bar{b}$ be two vectors such that $|\bar{b}|=1$ and $|\bar{b}\times\bar{a}|=3$. If $|\bar{b}\times\bar{a}-\bar{b}|^2=K$ then $\lim\limits_{x\to K^-}[x]=$ $\left([\bullet]is\,G.I.F\right)$

A)7 B) 8 C) 9 D) 10

59. Let $y=f(x)$ be a thrice differentiable function in $(-6,6)$.let the tangents to the curve $y=f(x)$ at (2,f(2)) and (3,f(3)) make angles $\frac{\pi}{6}$ and $\frac{\pi}{4}$ respectively with positive x-axis. If $9\sqrt{3}\int\limits_2^3\left\{\left(f'(t)\right)^2+1\right\}f''(t)dt=\alpha\sqrt{3}+\beta\ \ (\alpha,\beta\in Z)$, then $\lim\limits_{n\to\infty}\left(\frac{1}{\alpha+\beta}\right)^n=$ _____

A) 0 B) $\frac{1}{2}$ C) 1 D) $\frac{1}{8}$

60. If $5x+9=0$ is the directrix of the hyperbola $16x^2-9y^2=144$ and it's corresponding focus is S whereas S' is the other focus such that the circle having SS' as diameter is $x^2+y^2=K$, then K is

A) 16 B) 25 C) 9 D) 100

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61. Bag 'A' contains 4 blue and 9 red balls, whereas bag 'B' contains 3 blue and 10 red balls. One bag is selected at random and a ball is drawn from it. If it is known that the drawn ball is red and the probability that it is from bag 'A' is 'k', then k is

A) $\frac{10}{19}$ B) $\frac{9}{19}$ C) $\frac{9}{26}$ D) $\frac{10}{26}$

62. If [.] denotes the greatest integer function, then $\int_0^{\infty}\left[2e^{-x}\right]dx$ equals

A) 0 B) e^2 C) $2/e$ D) $\log 2$

63. If OABC is a tetrahedron such that $OA^2+BC^2=OB^2+CA^2=OC^2+AB^2$ then incorrect statement is

A) $OA\perp BC$ B) $OC\perp AB$ C) $OB\perp CA$ D) $AB\perp BC$

64. Let 'a' and 'b' be real constants such that the function f defined by

$f(x)=\begin{cases}x^2+ax+3, & x\le 1\\ 5x+b, & x>1\end{cases}$ be differentiable on R, then the value of $\int_{-1}^{1} f(x)\,dx=$ _____

A) $\frac{13}{9}$ B) $\frac{20}{3}$ C) $\frac{16}{3}$ D) $\frac{13}{6}$

65. If $\int\left(x^{12}+x^8+x^4\right)\left(2x^8+3x^4+6\right)^{1/4}dx=\frac{x^5}{a}\left(px^8+qx^4+r\right)^{b/4}+c$

Where c is constant of integration a,b,p,q,r are integers then $a+b-p+q-r=$

A) 35 B) 30 C) 25 D) 20

66. The value of $\sin\left(\frac{\pi}{900}\left\{\sum_{r=1}^{10}(r-\omega)\left(r-\omega^2\right)\right\}\right)$ is (where ω represents non real cube root of unity)

A) -1 B) 0 C) 1 D) $\frac{\sqrt{3}}{2}$

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67. The 3rd , 4th and 5th terms of $(x+a)^n, n \in N$ are respectively 60,160 and 240, then $\cos^{-1}\left(\frac{x}{a}\right)-\sin^{-1}\left(\frac{3}{n}\right)=$ ____

A) $\frac{\pi}{6}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{3}$ D) $\frac{\pi}{2}$

68. Let $A=\{x:[2\sin x]+[\cos x]=-3 \text{ and } x\in[0,2\pi]\}$ then the range of the function $f:A\to R$ defined by $f(x)=\sin x+\sqrt{3}\cos x$ is ____ $([\bullet] is\ G.I.F)$

A) $(-2,-1)$ B) $(-2,0)$ C) $(-1,-\frac{1}{2})$ D) $(-2,-\frac{1}{2})$

69. The value of $\int_{-\pi/2}^{\pi/2}\log\left(\frac{px^2+qx+r}{px^2-qx+r}(a+b)|\sin x|\right)dx$

A) $\frac{\pi}{2}\log\left(\frac{a+b}{2}\right)$ B) $\pi\log\left(\frac{a+b}{2}\right)$ C) $\frac{\pi}{4}\log\left(\frac{a+b}{2}\right)$ D) $\frac{\pi}{8}\log\left(\frac{a+b}{2}\right)$

70. If $a\sin\theta=b\sin\left(\theta+\frac{2\pi}{3}\right)=c\sin\left(\theta+\frac{4\pi}{3}\right)\neq 0, \theta\in(0,2\pi)$ and $M=\begin{pmatrix} bc & 0 & 0 \\ 0 & ca & 0 \\ 0 & 0 & ab \end{pmatrix}$,

then the CORRECT statement(s) among the following, is/are

i) trace of M is equal to zero ii) $\sin^{-1}(ab+bc+ca)=0$

iii) $\cos^{-1}\{abc(ab+bc+ca)\}=1$

A) only i and ii are correct B) only ii and iii are correct

C) only i and iii are correct D) i,ii and iii are correct

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. For a certain curve $y=f(x)$ satisfying $\frac{d^2y}{dx^2}=6x-4, f(x)$ *has a* local minimum value 5 when $x=1$. If the equation of the curve is $y=x^3-Kx^2+x+L$ and the global maximum and global minimum values of f(x), given that $0\le x\le 2$ are $'\alpha'$ and $'\beta'$, then $\alpha+\beta-K-L=$ ___

72. If the area of the region bounded by the straight lines $x=0, x=2$ and the curves $y=2^x, y=2x-x^2$ is $\left(\frac{K}{\ln 2}-\frac{L}{3}\right)$, then the product of 'K'and 'L'is._____

73. Let $S_n=1+q+q^2+...+q^n$

$T_n=1+\left(\frac{q+1}{2}\right)+\left(\frac{q+1}{2}\right)^2+....+\left(\frac{q+1}{2}\right)^n$,

Where q is a real number and $q\neq 1$.

If ${}^{101}C_1+{}^{101}C_2\times S_1+....+{}^{101}C_{101}\times S_{100}=\alpha^k T_{100}$ and α is prime number then $\alpha+k$ is equal to _____.

74. The number of symmetric relations defined on the set $\{5,7,8,9\}$ which are not reflexive , is____

75. If $\sum_{i=1}^{9}(x_i-10)=9$ and $\sum_{i=1}^{9}(x_i-10)^2=45$ and the standard deviation of $x_1,x_2,x_3,.....,x_9$ is K, then $[K]=$ _____ $([\bullet] is\ G,I,F)$

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